

June 17, 2026

To Whom It May Concern,

I am writing to express my strong interest in the PhD position "Integrating Quantum and Classical Sensors for Long-Duration Inertial Navigation" at the University of Southampton. As an AI Data Science Engineer with a passion for innovative solutions, I believe that this project aligns perfectly with my skills and expertise.

My experience in developing a linear-time state space model (Medical_MambaTS) has allowed me to appreciate latency inference in various applications. In the context of inertial navigation, Medical_MambaTS 's ability to process millisecond bedside inference can be adapted to fusion algorithms for real-time navigation corrections. I am

Furthermore, my experience in building a Physics-Informed Operator Network (PI_DeepONet) using *PyTorch Autograd* has provided me with a deep understanding of physical constraints and

I am confident that my skills and experience make me an ideal candidate for this project. I look forward to the opportunity to contribute to the development of hybrid quantum-classical navigation systems and explore their applications in GPS-denied environments.

Sincerely,